SPE-230098-MS

A Deep-To-Shallow Seasonal Energy Storage Strategy to Improve Geothermal Longevity in Regina, Saskatchewan

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Sedimentary basins, both deep and shallow, offer valuable low-carbon energy solutions by serving as significant geothermal energy sources. In cold climates like the Canadian prairie provinces, extracting geothermal energy from low-enthalpy subsurface systems provides a practical way to meet seasonal heat demands. This study focuses on optimizing geothermal energy utilization through deep-to-shallow thermal energy storage strategies, enhancing system sustainability, and supporting long-term energy security.

In this study, we built a numerical model of three aquifers, namely Winnipeg, Deadwood, and Mannville. These formations are key geothermal resources and have provided high permeability and suitable temperatures for direct heat use since 1978. The shallower Mannville Aquifer, at around 850 m, consists of interbedded sandstone and shale with regional groundwater flow. Two doublet systems are incorporated to evaluate deep-to-shallow aquifer thermal energy transfer. Seasonal energy demand is evaluated using data from a University of Regina residential facility. The methodology involves simulating fluid flow, heat transfer, and aquifer interactions to optimize extraction rates while minimizing thermal breakthroughs.

Numerical simulations indicate that integrating a deep-to-shallow thermal energy storage approach significantly enhances geothermal system efficiency. The highest cumulative production rate from both aquifers was set at 2400 m^3/day. During low-energy demand periods, deep geothermal well production could be reduced to a constant rate of 1000 m^3/day, with excess energy stored by injecting it into the shallower geothermal reservoir. Storing excess energy from deep aquifers in the shallower Mannville aquifer during low-demand periods slows thermal depletion of the deep reservoir, preserving higher temperatures for extended use. Observations show that seasonal energy storage improves temperature stability, reduces thermal breakthroughs, and optimizes extraction rates. Additionally, findings emphasize the importance of managing production patterns to align with seasonal heat demand, ensuring long-term sustainability. The study concludes that implementing Aquifer Thermal Energy Storage (ATES) enhances geothermal system longevity, increases energy efficiency, and provides a sustainable solution for regions

like Saskatchewan. Overall, the deep-to-shallow thermal exchange method offers an effective strategy for optimizing geothermal energy utilization while improving thermal stability in subsurface reservoirs.

Our modeling approach assesses the novel deep-to-shallow geothermal storage strategy to optimize energy extraction and seasonal heat management. The results provide insights into enhancing energy efficiency, increasing geothermal system sustainability, and improving thermal storage in the shallower aquifer during low-demand months.

Introduction

Rising global population and energy demand are intensifying climate change, prompting a worldwide shift toward clean and renewable energy sources such as solar, wind, and geothermal (Beernink et al., 2024; Li et al., 2022). The burning of fossil fuels has been a major contributor to greenhouse gas (GHG) emissions. In contrast, geothermal energy—derived from the Earth's interior—is a reliable, clean alternative that holds great promise for meeting energy needs, particularly in regions like the Canadian prairie provinces, where heating and cooling account for 80% of total energy demand (Barbier, 2002; Grasby et al., 2009). Sedimentary basins are suitable for low-to-moderate enthalpy geothermal applications, especially for communities situated above such formations (Majorowicz & Grasby, 2021). The Western Canadian Sedimentary Basin (WCSB), which spans northern British Columbia, northwest Northwest Territories, Alberta, southern Saskatchewan, and southwestern Manitoba, overlays Precambrian crystalline rocks and reaches sedimentary thicknesses of up to 6 km near the Cordillera, thinning toward the Canadian Shield. This geological structure plays a pivotal role in determining geothermal gradients, which vary significantly depending on location: from as low as 20 °C/km in areas distant from plate boundaries to moderate values of 30–45 °C/km in southern Alberta and higher values of 40–50 °C/km in the northwestern part of the basin. Temperature data from oil and gas wells indicate that drilling to depths of 2–3 km yields reservoirs with temperatures ranging from 60 to 100 °C (Jessop & Vigrass, 1989; Majorowicz & Grasby, 2021; Rangriz Shokri et al., 2021; Vigrass et al., 2007). Unlike intermittent sources such as solar and wind, which are weather-dependent, geothermal energy offers a stable supply, though sustainable extraction requires careful management (Hau et al., 2020). Matching production with periodic demand is critical, but regulating flow rates can damage reservoirs by reducing permeability, causing cold water breakthroughs, and triggering salt precipitation due to geochemical changes. To address these challenges, integrating Aquifer Thermal Energy Storage (ATES) into geothermal systems provides a promising solution. High-temperature geothermal systems (>60 °C) serve as the main heat source, while low-temperature ATES (<30 °C) functions as a storage element. These low-temperature TES systems are especially widespread in the Netherlands, where over 2,500 installations support both heating and cooling needs (Daniilidis et al., 2017, 2022; Voskov et al., 2024).

Geological Background of Studied Area

Studies on Saskatchewan geology have identified seven major aquifer systems within the Williston Basin, intermittently separated by regional aquitards (Movahedzadeh et al., 2021). These aquifers include the deepest Basal Aquifer, ascending to near-surface systems, with one of the uppermost aquifers of the Tertiary period recognized as the Mannville Aquifers from the Cretaceous era (Ufondu, 2017a). During the winter of 1978–79, a geothermal well was drilled on the University of Regina (U of R) campus in Saskatchewan. The project aimed to withdraw hot water to supply a planned large sports complex on campus. Valuable geological and geothermal data were obtained from the U of R well. The well reached a depth of 2,226 m and was completed in the Winnipeg and Deadwood formations, with an open hole section from 2,034 to 2,226 m.

The lithology of the sedimentary succession at the site is categorized into three broad units: the Basal Clastic Unit, the Carbonate–Evaporite Unit, and the Upper Clastic Unit, spanning ages from the Cambrian to the Quaternary. The Cambrian-aged Deadwood Formation overlies the Precambrian crystalline basement and underlies the Middle Ordovician Winnipeg Formation. The Deadwood Formation is generally composed of coarse sandstone with an upper shale layer, while the lower Winnipeg Formation features fine to medium-grained sandstone known as the Black Island Member, which is capped by a shaley Icebox layer. The stratigraphic tops of these formations are depicted in Figure 1.

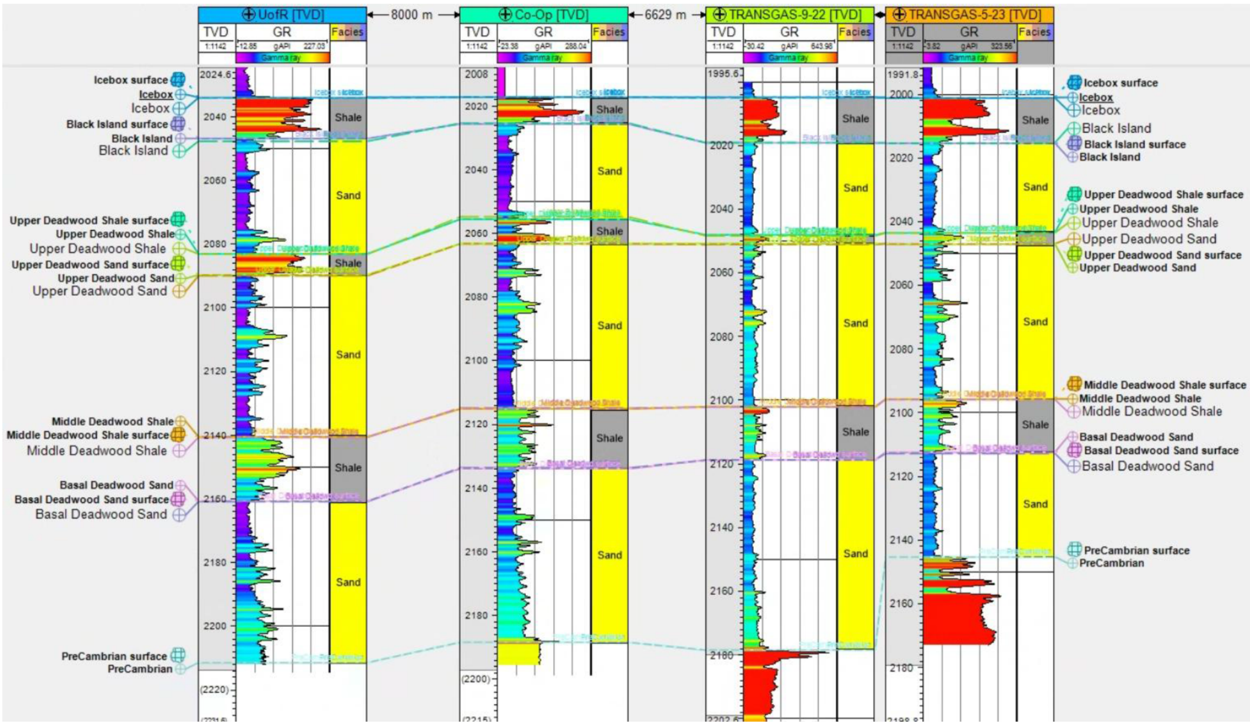

Figure 1—Well section and identified formation tips for Winnipeg and Deadwood as the deep aquifer

The Mannville group in WCSB, shown in well section view of the studied wells at the target location in Figure 2, is found in the depth range of 600-1200 m with a maximum thickness of approximately 150 m in southeastern Saskatchewan. It is primarily composed of sandstone with interbedded shale. This formation has been mostly used as a disposal zone by oil and gas industry. The Mannville aquifer exhibits the highest permeability within the Mesozoic group, reaching up to 10^{-8} m^2. It is confined by the Colorado-Lea Park and Vanguard shaly units, which act as aquitards from the top and bottom, respectively (Jellicoe, 2020; Woroniuk, 2023).

3D geological models of the Deadwood-Winnipeg and Mannville aquifers were built to study the performance of deep and shallow geothermal reservoirs. These models were constructed by integrating geological data collected from drilled wells within the target area. The dataset includes core analysis, log curves interpretation, and empirical correlation, such as those between porosity and permeability, as well as thermal and physical properties of both fluids and rock within the aquifers. Reservoir simulations show the efficiency of deep geothermal reservoirs to supply energy during periods of high demand, and to store surplus energy in the low demand seasons. These results provide a deeper understanding of variations in parameters such as temperature and pressure at the borehole and surface levels, along with the amount of energy harvested and stored from shallow and deep geothermal aquifers according to seasonal production patterns.

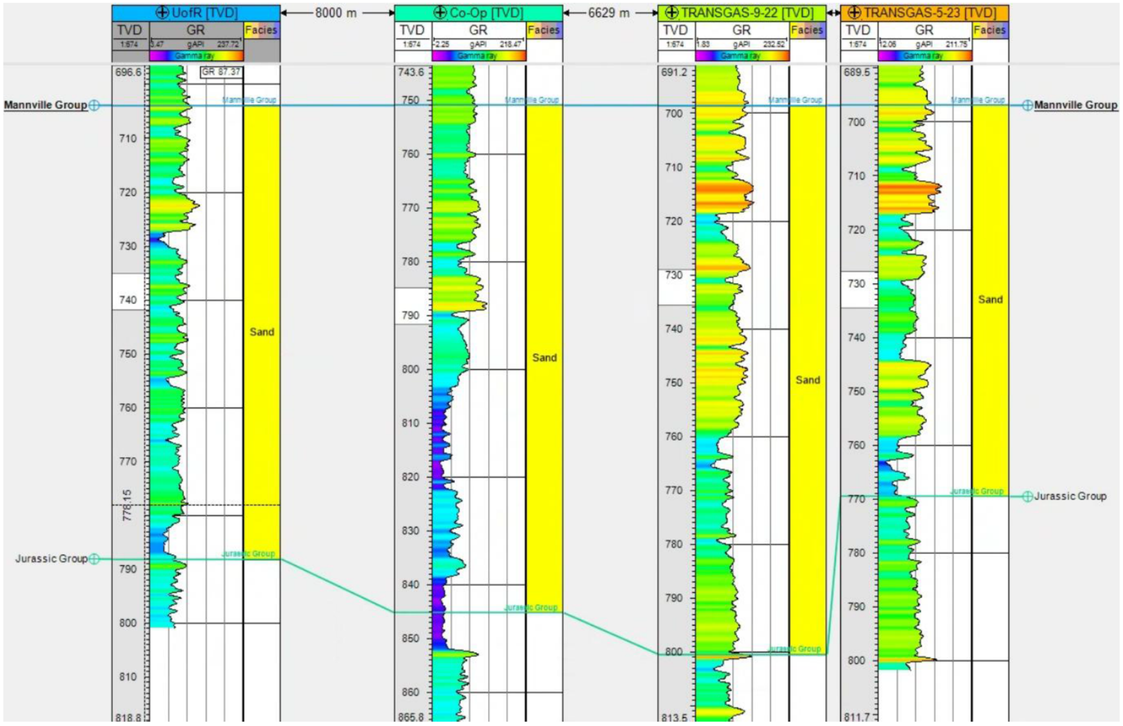

Figure 2—Well section and identified formation tips for Mannville Group

Geological Modelling

The geological setting at the City of Regina area, Saskatchewan, was studied using available logs of four drilled wells within or near the target site. Gamma ray (GR) logs were employed to identify stratigraphic horizons and formation tops, as illustrated in the well section of Figure 1. Detailed well information, including wellhead location, measured depth (MD), true vertical depth (TVD), and Kelly bushing (KB), is summarized in Table 1.

Table 1—Detailed information of existing wells in the studied area

Well Name	Well ID	KB (m)	TVD (m)	MD (m)
UofR	131/03-08-017-19W2/02	580.9	2225.6	2225.6
Co-Op	191/04-04-018-19W2	594.5	2198.5	2376
TRANSGAS-9-22	101/05-23-18-19W2	612.8	2232.0	2232.0
TRANSGAS-5-23	101/09-22-018-19W2	610.4	2195.0	2195.0

The depths to the top of the Deadwood and Winnipeg formations, representing the deep geothermal reservoirs, as well as the depths to the top of the Mannville and Jurassic groups, representing the shallow geothermal aquifer, in these four studied wells are listed in Tables 2 and 3.

Table 2—Formations top depth for the Winnipeg and Deadwood-deep reservoir

Well/Surface	Icebox	Black Island	Upper Deadwood Shale	Upper Deadwood Sand	Middle Deadwood Shale	Basal Deadwood Sand	Precambrian
UofR	2034.01	2047.65	2083.10	2089.91	2140.63	2160.81	2211.39
Co-Op	2197.53	2205.71	2235.16	2243.62	2295.15	2313.97	2188.36
TRANSGAS-9-22	2004.95	2019.10	2048.32	2051.09	2102.14	2118.75	2178.40
TRANSGAS-5-23	2001.15	2015.30	2043.60	2046.98	2095.89	2112.80	2445.29
Average Net Thickness	12.53	30.61	5.36	50.55	18.13	79.28	

Table 3—Formations top depth for Mannville Group- shallow reservoir

Well/Surface	Mannville Group	Jurassic Group
UofR	703.73	788.14
Co-Op	754.80	852.01
TRANSGAS-9-22	896.72	800.67
TRANSGAS-5-23	896.72	769.28

The identification of top formation horizons facilitated the creation of structure maps for each formation, as shown in Figures 3 and 4, along with the location of the wells. A 3D geological model was built in Computer Modelling Group (CMG) software for the area surrounding the City of Regina, covering a region of 25 km × 15 km as indicated in Figure 5. The model represents both deep and shallow reservoirs, with a horizontal grid size of 500 m × 500 m. In the vertical (z) direction, the Winnipeg and Deadwood Formations were each separately divided into 10 layers.

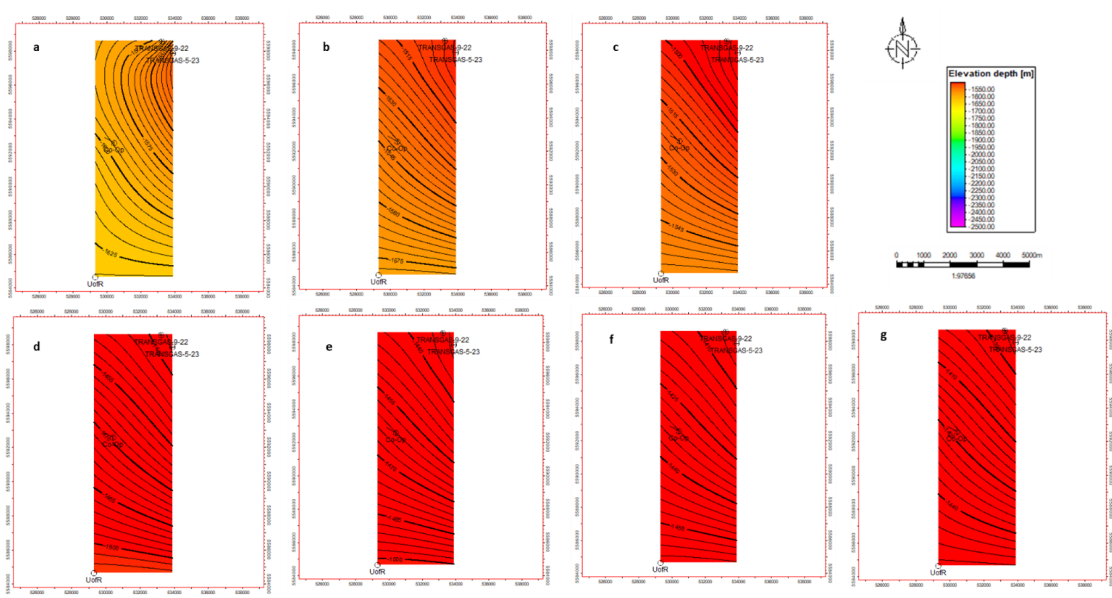

Figure 3—Structure map respectively a) Precambrian basement, b) Basal Deadwood Sand, c) Middle Deadwood Shale, d) Upper Deadwood Sand, e) Upper Deadwood Shale, f) Black Island, g) Icebox

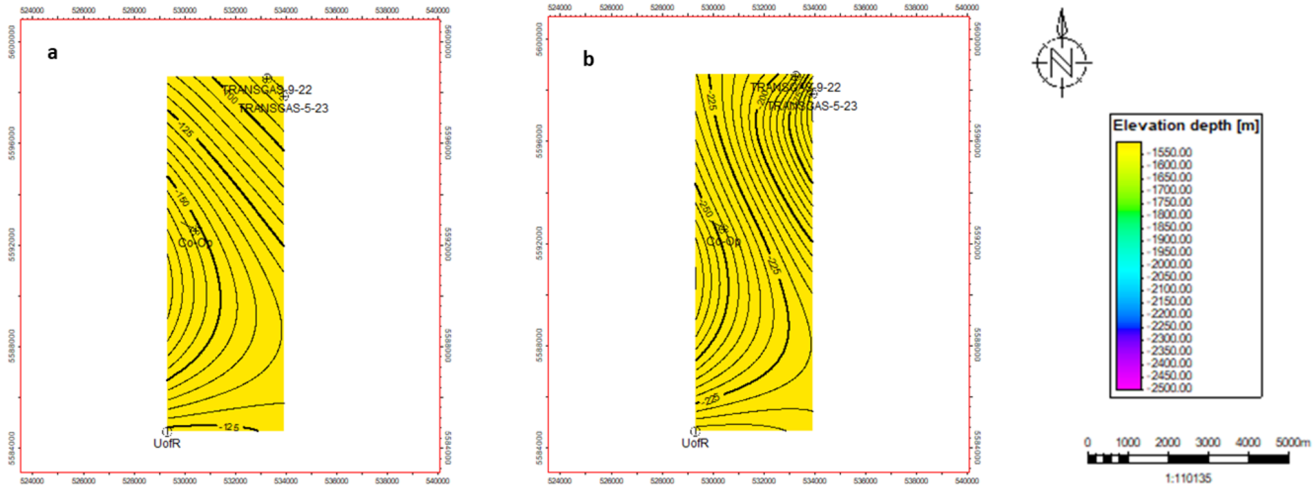

Figure 4—Structure map respectively a) Mannville Group, b) Jurassic Group

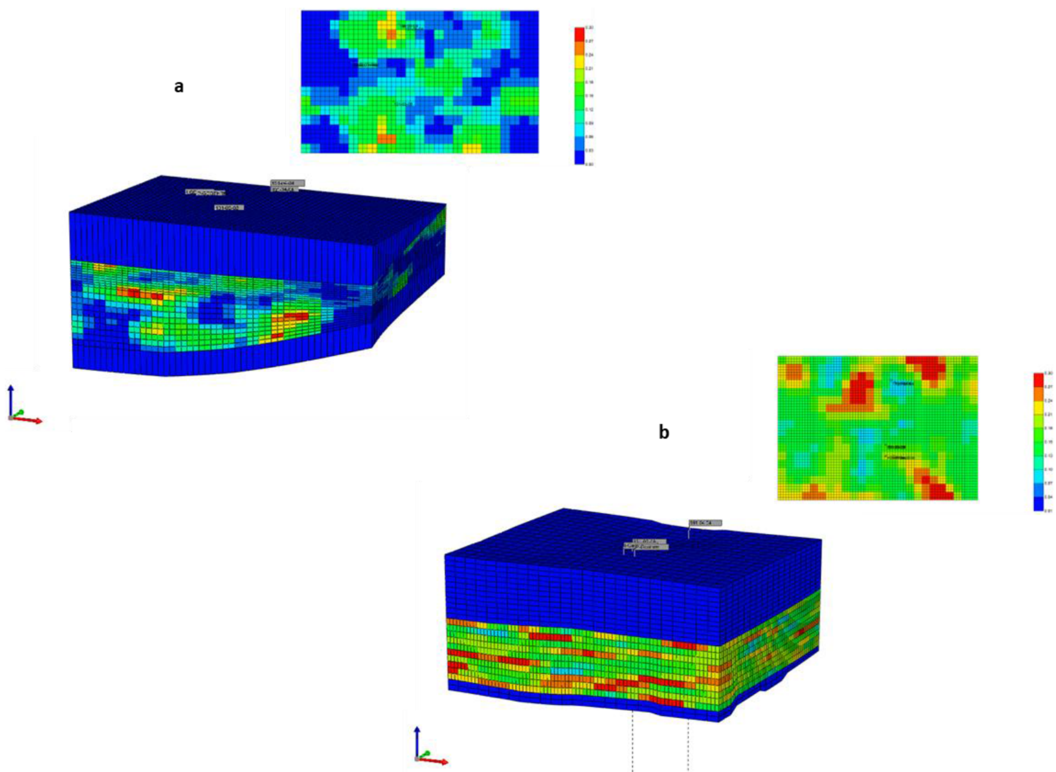

Figure 5—3D Reservoir models for a) Deadwood-Winnipeg Formations, b) Mannville along with top view to a layer porosity distribution

Information reported by Vigrass et al. (2007) from the University of Regina geothermal well provided invaluable insight into the downhole temperature and geothermal gradient at the site. A temperature log, conducted 1729 days after completion of the drilling process, supported the integration of a thermal distribution into the model based on a geothermal gradient of 26 °C/km.

Density-porosity sandstone logs from the UofR and Co-Op were utilized to simulate porosity distributions within the model grid for both shallow and deep reservoirs. Permeability data from Nesheim (2021) and Ufondu (2017) were incorporated to establish empirical correlations between permeability (y) and porosity (x) for the Deadwood and Winnipeg Formations, respectively, as follows:

$y = 1399\ x^{3.343}$

$y = 51.081\ x^{-0.451}$

A limited number of laboratory-measured permeability data points were also incorporated into the permeability correction and geomodel calculations. The characteristic thermal and physical properties of the reservoir rock and saturated fluids, including thermal conductivity, heat capacity, compressibility and density, were gathered from existing studies on the Saskatchewan Basal Clastics by Ferguson & Grasby (2014) and Ufondu (2017) as well as on water injection in southeastern Saskatchewan by Jellicoe (2020). Additional data on drainage relative permeabilities across various rock types were sourced from Rangriz Shokri et al. (2019) who reviewed previous studies and compiled datasets used in the Aquistore CO_2 storage modelling. The Mannville Formation's relative permeability curves were interpreted based on work conducted by Wang et al. (2010) on oil production in southeastern Saskatchewan.

Aquifer water production and injection are conducted through an open-loop system using a doublet well configuration. For the deep aquifer, the two wells are spaced 5 m apart at the surface, with the deviation kick-off point beginning at a depth of 500 m. This directional drilling design results in a bottomhole separation of approximately 1200 m at a depth of around 2200 m. Such spacing at depth is intended to reduce the risk of early thermal breakthrough and extend the operational lifespan of the geothermal system.

In contrast, the doublet system for the Mannville reservoir is a set of two vertical wells, drilled to a depth of 893 m. To ensure sealing integrity, the overburden and underburden layers surrounding both reservoirs were modeled using shaly units in accordance with the southeastern Saskatchewan lithological characteristics.

A reference temperature of 16 °C was used for the simulation calculation. Based on the estimated hydraulic pressures within the reservoirs, both the pump placement and required pump power were calculated. Selected operating parameters are summarized in Table 4.

Table 4—wellbore and pump information for the two geomodels

Parameters/Aquifers	Deadwood-Winnipeg	Mannville
Wellbore length (m)	2571	893.6
Pump depth (m)	798	893.6
Pump power (kW)	230	260

Estimation of Heat Demand in a Case Study

Retrofitting renewable energy systems, such as geothermal, into an already constructed building requires the knowledge of that infrastructure's energy demand. This study aimed to better understand the energy needs of a typical facility located in the studied geographic region, characterized by its unique weather conditions. The energy demand curve serves as a foundation for designing precise patterns of production, storage and injection.

The study accessed the energy demand curve of a facility located at the University of Regina campus, illustrated in Figure 6. This 14-storey building currently relies on natural gas for heating, which is distributed throughout the structure via a radiant system to the building.

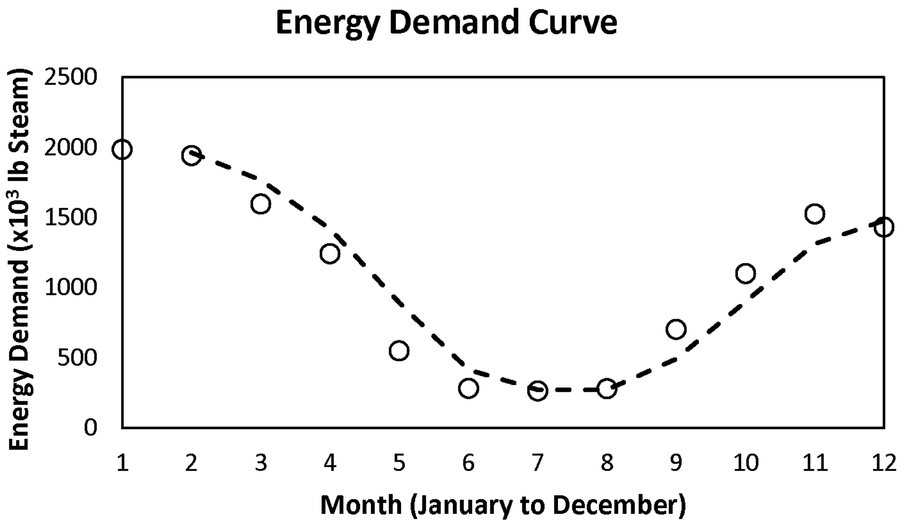

Figure 6—Energy demand vs month of operation for a facility at the University of Regina Campus, Saskatchewan

According to the reported energy curve, the first three months of the year show the highest heat demand, while the summer months (June, July and August) exhibit minimal heating requirements. As previously mentioned, the study focused on optimizing the lifespan of the project by managing heat flow and adhering to the seasonal demand pattern. This strategy enables production at a relatively constant flow rate while allowing surplus thermal energy to be stored in more accessible aquifers.

Simulation Scenarios

Three scenarios were considered in this study to understand how different fluid withdrawal strategies from geothermal reservoirs affect key aquifer parameters such as temperature, pressure (both at the bottomhole and surface), and the extent of cold temperature plumes within the injection zones of both reservoirs.

The scenarios include:

- constant production from each aquifer independently at a rate of 2400 m^3/day,
- adjusting production based on the energy demand of the facility,
- maintaining a relatively constant production rate from the deep reservoir while storing surplus energy in the shallow aquifer.

Table 5 shows the production and injection rates (in m^3/day) for scenarios 2 and 3, which incorporate shallow aquifers into deep geothermal operations by aligning with the monthly energy requirements of the case-studied building. Changes in the key parameters were observed over the operational period, spanning from 2024 through the end of 2050.

Table 5—Production and injection rates in m3/day for scenario 2 and 3 to the associated aquifers

Month	Scenario 2		Scenario 3			
	Production Rate	Production Rate	Production Rate (Deep Aquifer)	Injection Rate (Deep Aquifer)	Production Rate (Shallow Aquifer)	Injection Rate (Shallow Aquifer)
Jan	2400	2400	2400	2400	Shut-in	Shut-in
Feb	2400	2400	2400	2400	Shut-in	Shut-in
Mar	2000	2000	1000	1000	1000	1000
Apr	1500	1500	1000	1000	500	500
May	700	700	1000	700	Shut-in	300
Jun	Shut-in	Shut-in	1000	Shut-in	Shut-in	1000
Jul	Shut-in	Shut-in	1000	Shut-in	Shut-in	1000
Aug	Shut-in	Shut-in	1000	Shut-in	Shut-in	1000
Sep	900	900	1000	900	Shut-in	100
Oct	1400	1400	1000	1000	400	400
Nov	1900	1900	1000	1000	900	900
Dec	1800	1800	1000	1000	800	800

Results and Discussion

The heat demand during the first three months of the year, typically the period with the highest energy requirements, was compared against the thermal energy that can be harvested from both shallow and deep reservoirs at a constant production rate of 2400 m^3/day. The results indicate that the deep reservoir can supply about 6 times the building's required heat, while the shallow reservoir can produce roughly double the demand. The comparative values of the energy demand vs. potential harvested heat are presented in Table 6.

Table 6—The energy requirement for the first three months of a year for the studied case comparing to calculated potential heat from two aquifers

Month (peaks)	Steam (lb)	bBTU heat from Steam	bBTU/month Shallow	bBTU/month Deep
January	1985527.68	2.371	4.129	13.076
February	1940606.4	2.317	4.129	13.079
March	1596670.08	1.906	3.44	10.901

Figure 7 illustrates the variation in bottomhole temperature over a 25-year operational period. Under the highest constant production of 2400 m^3/day, the temperature declined from slightly from 61.11 °C to 60.89 °C. However, aligning production with actual energy demand helped maintain a higher reservoir temperature of 61.0 °C. Interestingly, reducing the production rate to 1000 m^3/day led to a minor increase in bottomhole temperature, suggesting enhanced thermal sustainability for the deep reservoir.

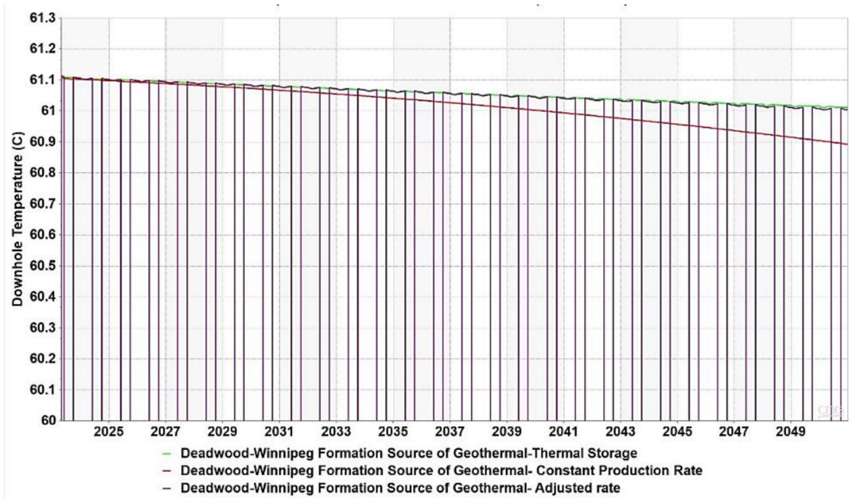

Figure 7—Bottomhole temperature in the three studied scenarios for the deep reservoir

In contrast, temperature dynamics within the shallow aquifer are more pronounced, as depicted in Figure 8. Constant production reduced the reservoir temperature from 32.88 °C to 32.16 °C over 25 years. However, storing surplus heat from the deep reservoir into the shallow aquifer mitigated this decline, ultimately raising the reservoir temperature to 32.85 °C by the end of the operational period.

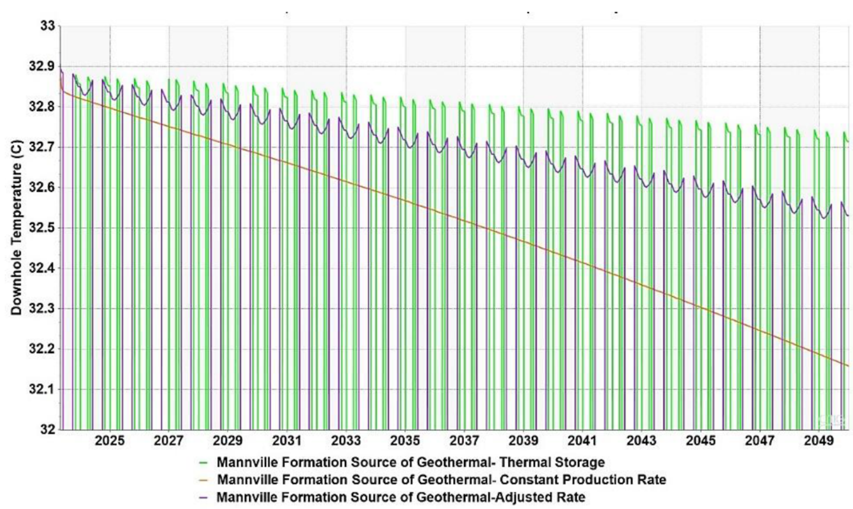

Figure 8—Bottomhole temperature in the three studied scenarios for the shallow reservoir

As shown in Figure 9, the surface temperature of the produced fluid from the deep and shallow aquifers were measured at 54.46 °C and 30.701 °C, respectively. Adjusting the production rate to match the facility's energy demand resulted in a slight reduction in surface temperature, to 54.29 °C for the deep aquifer and 30.376 °C for the shallow aquifer. However, integrating shallow storage with deep reservoir production helped prevent a significant temperature decline in the shallow reservoir, while having minimal effect on the surface production temperature from the deep reservoir. These results can be seen in Figure 10.

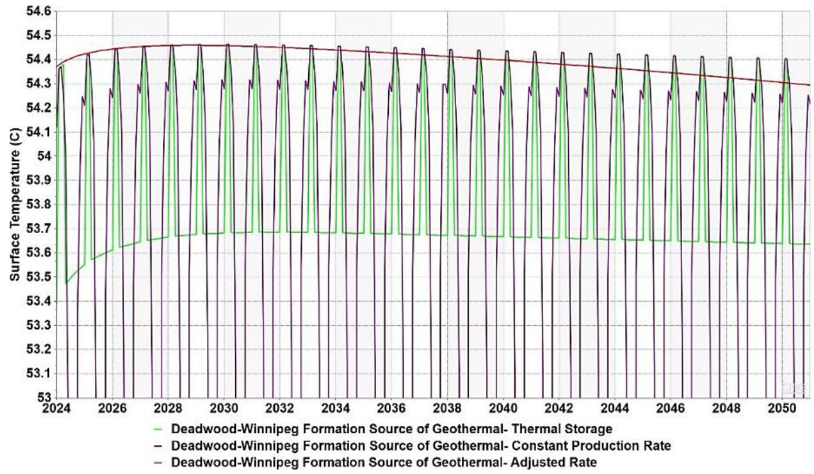

Figure 9—Surface temperature in the three studied scenarios for the deep reservoir

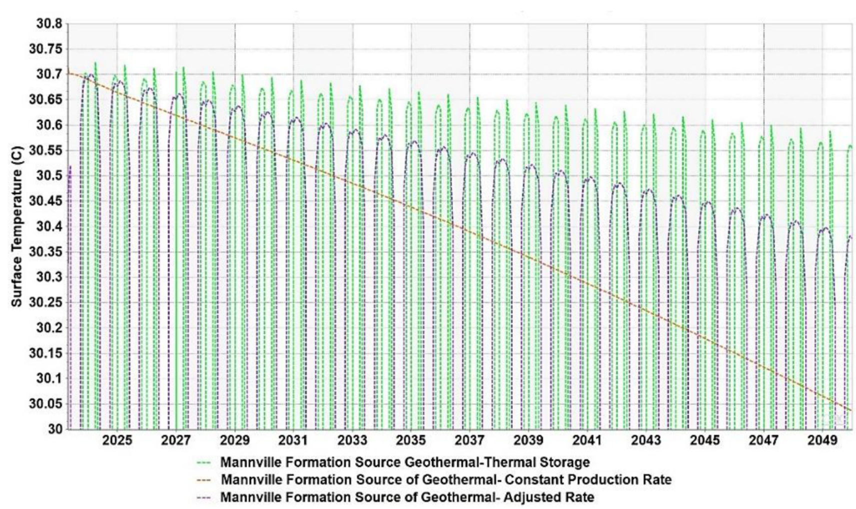

Figure 10—Surface temperature in the three studied scenarios for the shallow reservoir

Figures 11 and 12 illustrate that optimizing heat extraction from geothermal reservoirs, via the evaluated heat management scenarios, can effectively mitigate the dispersion of cold plumes further into the reservoir. Strategic regulation of heat harvesting enhances the reservoir's longevity and reduces the risk of thermal breakthrough incidents.

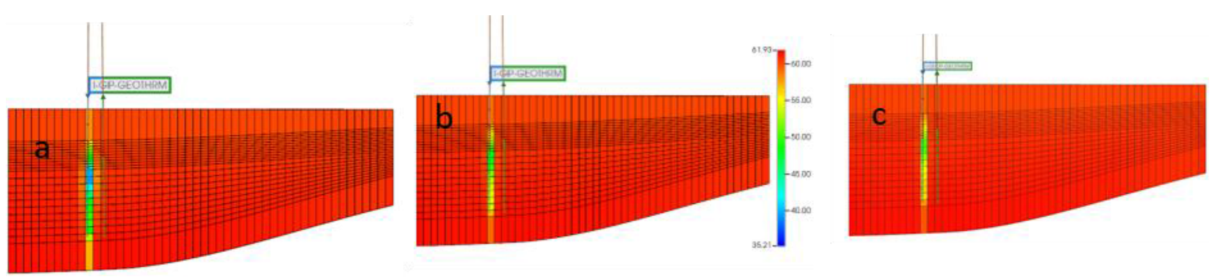

Figure 11—Temperature Distribution for Winnipeg-Deadwood Geothermal Reservoir vertical view a) Constant Production Rate, b) Adjusted Rate, c) Thermal Storage

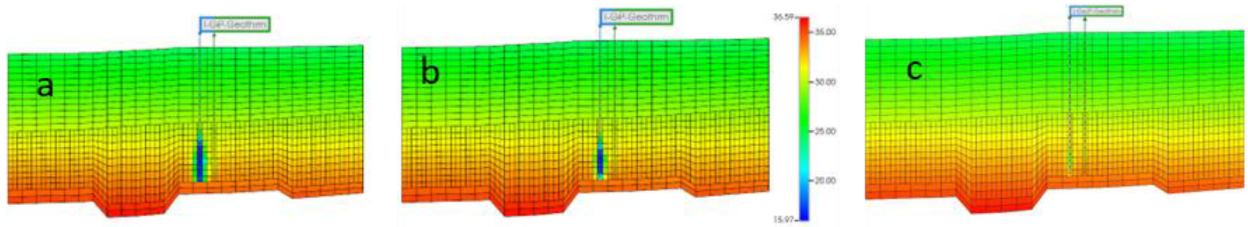

Figure 12—Temperature Distribution for Mannville Geothermal Reservoir vertical view a) Constant Production Rate, b) Adjusted Rate, c) Thermal Storage

Conclusions

This study highlights the critical role of geothermal energy in meeting local energy demands sustainably, with direct heat potentially supplying up to 80% of community needs. The Western Canadian Sedimentary Basin presents a promising geothermal resource, especially in Saskatchewan, with aquifers at varying depths offering distinct thermal profiles. By examining both the Mannville Aquifer and the deeper Basal Clastic Unit, we investigated operational strategies to optimize heat extraction while preserving reservoir integrity.

A key challenge in extracting hot water from deep geothermal reservoirs is maintaining high reservoir temperatures without inducing thermal breakthrough or significant temperature decline. To ensure the longevity of deep geothermal systems, it is essential to align energy production with the demand profile of the supply facility. However, fluctuating production rates from the deep aquifer can negatively affect reservoir permeability due to geochemical changes. Therefore, a constant production rate is recommended, with surplus heat stored in shallower aquifers.

In conclusion, strategic integration of shallow and deep aquifers, guided by facility demand profiles, offers a resilient and sustainable model for geothermal energy production in Western Canada.

Acknowledgments

We also acknowledge Schlumberger for supplying the Petrel software package, and Computer Modelling Group Ltd. (CMG) for providing the CMG software suite. Authors of this paper would like to extend their sincere appreciation to Saskatchewan Geological Survey and geoLOGIC Systems Ltd. for their persistent support in providing geological data.

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